EMS LAB 1

```
clear;
clc;
close all;
```

1. NTC PARAMETERS

```
R0   = 10000;
beta = 3380;
T0   = 25;

Tmin = 20;
Tmax = 50;

Vdd = 3.3;
```

2. TEMPERATURE

```
T0K = T0 + 273.15;

T = Tmin:0.1:Tmax;      % Temperature vector, step = 0.1 °C
TK = T + 273.15;
```

3. NTC MODEL

```
Rntc = R0 .* exp(beta .* (1./TK - 1/T0K));

% For an NTC:
% Tmin -> maximum resistance
% Tmax -> minimum resistance
Rmax = Rntc(1);         % First value corresponds to Tmin
Rmin = Rntc(end);       % Last value corresponds to Tmax
```

4. DIMENSION R1

```
%R1 = sqrt(Rmax * Rmin);

R1 = 10000;

fprintf('R1 ideal = %.2f kOhm\n', R1/1000);
```

```
R1 ideal = 10.00 kOhm
```

5. VOLTAGE DIVIDER

```
%
%       Vdd
%        |
%       NTC
%        |
```

```
%         +------ Vin
%         |
%         R1
%         |
%        GND

Vin = Vdd .* R1 ./ (Rntc + R1);


fprintf('\n--- Vin ---\n');
```

```
--- Vin ---
```

```
fprintf('Vin min = %.3f V\n', min(Vin));
```

```
Vin min = 1.491 V
```

```
fprintf('Vin max = %.3f V\n', max(Vin));
```

```
Vin max = 2.330 V
```

6. CURRENT

```
I = Vdd ./ (R1 + Rntc);


fprintf('\n--- Current ---\n');
```

```
--- Current ---
```

```
fprintf('I max = %.3f mA\n', max(I)*1000);
```

```
I max = 0.233 mA
```

7. NTC POWER

```
Pntc = I.^2 .* Rntc;


fprintf('\n--- NTC Power ---\n');
```

```
--- NTC Power ---
```

```
fprintf('P max = %.3f mW\n', max(Pntc)*1000);
```

```
P max = 0.272 mW
```

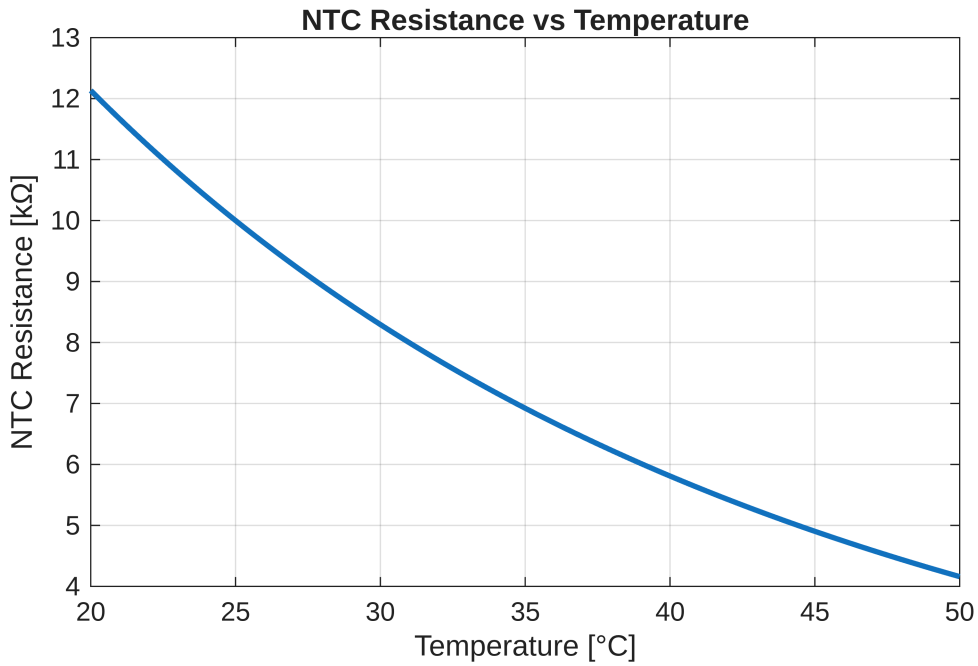
8. GRAPHS

```
figure;

plot(T,Rntc/1000,'LineWidth',2);

grid on;
```

```
xlabel('Temperature [°C]');
ylabel('NTC Resistance [k\Omega]');
title('NTC Resistance vs Temperature');
```

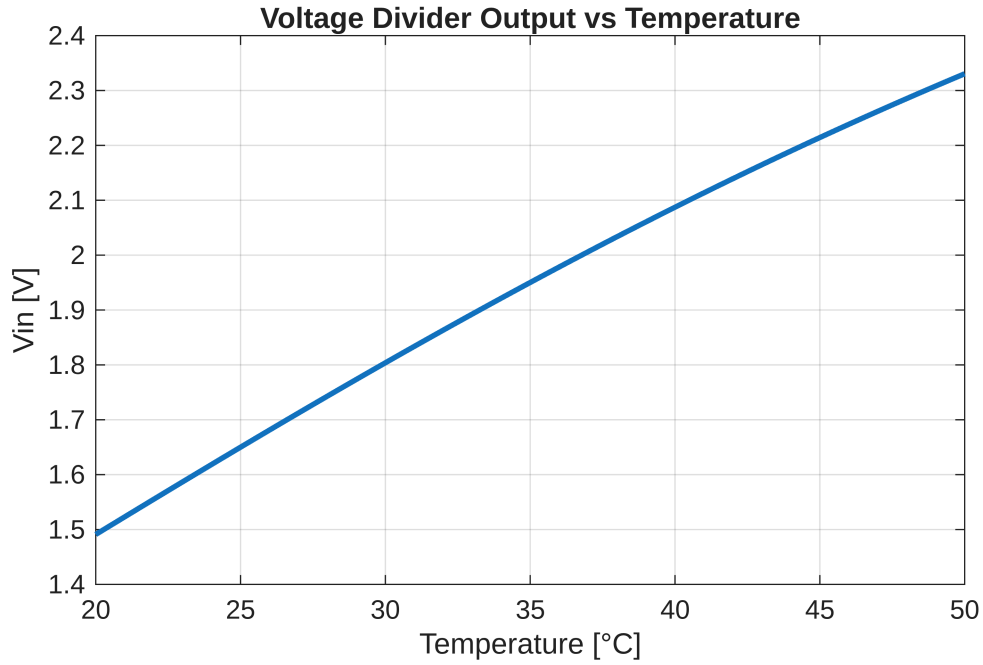



```
figure;

plot(T,Vin,'LineWidth',2);

grid on;

xlabel('Temperature [°C]');
ylabel('Vin [V]');
title('Voltage Divider Output vs Temperature');
```

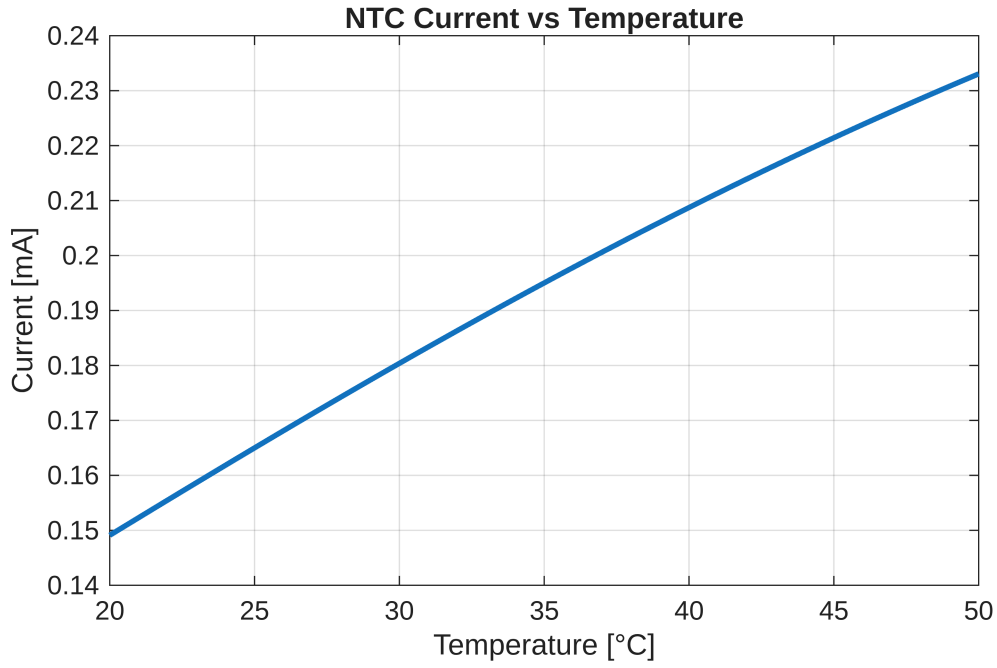



```
figure;

plot(T,I*1000,'LineWidth',2);

grid on;

xlabel('Temperature [°C]');
ylabel('Current [mA]');
title('NTC Current vs Temperature');
```



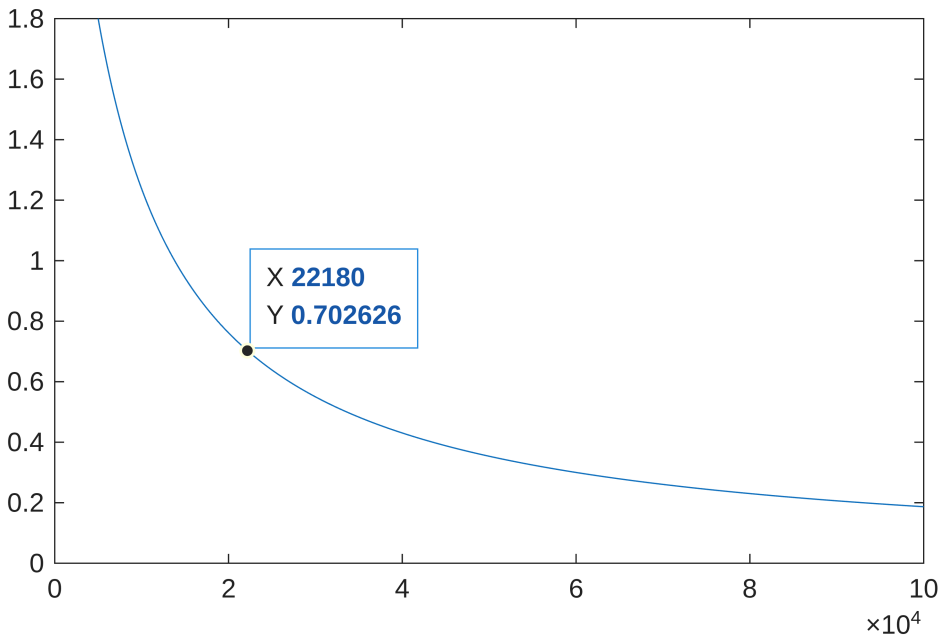
Soil Sensor

```
Rfixa = (5*10 ^3 : 10 : 100 * 10 ^3);

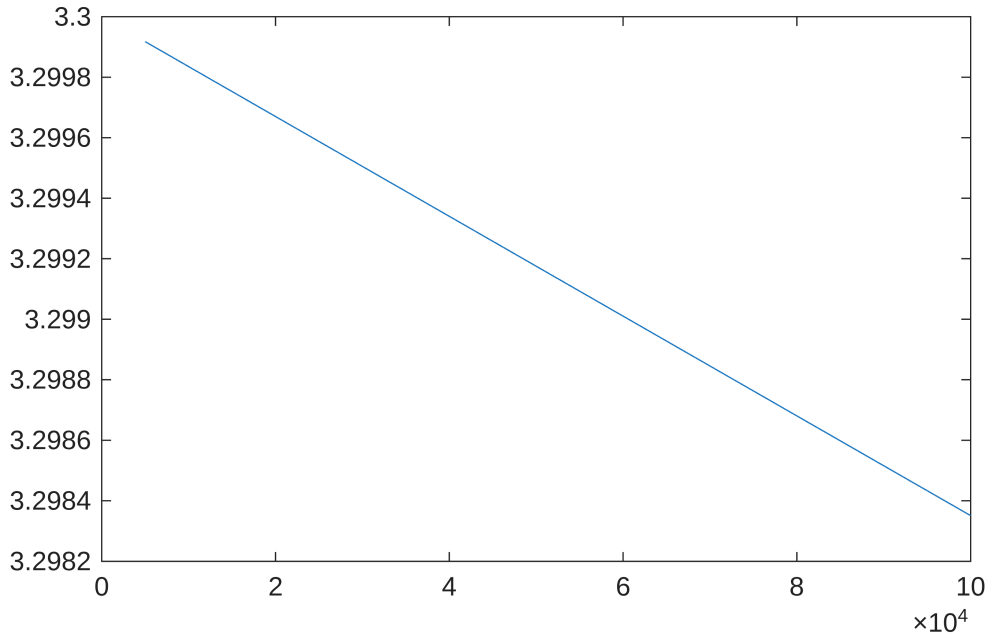
Rmax = 200 * 10^6;
Rmin = 6 * 10 ^3 ;

voutdry  = 3.3 .* (Rmax ./ (Rmax + Rfixa));
voutwet  = 3.3 .* (Rmin ./ (Rmin + Rfixa));
plot (Rfixa, voutwet);

ax = gca;
chart = ax.Children(1);
datatip(chart,2.218e+04,0.7026);
```




```
plot (Rfixa, voutdry);
```




```
adcM= 3.3*4095 / 3.3
```

```
adcM =
4095
```

```
adcm = 0.7026 * 4095 / 3.3
```

```
adcm =
871.8627
```